

Table 5: Comparison of the quantum resources required for inversion

n	Source	#Toffoli	#CNOT	Width	Toffoli depth	Full depth	DW-cost
163	[JSB+25]	39483	541839	48420	9	1362	65948040
	Ours (Karatsuba)	39483	500772	57194	8	708	40493352
	Ours (AC-based)	8154	1761213	11941	8	6058	72338578
233	[JSB+25]	63230	972615	76109	10	1784	135778456
	Ours (Karatsuba)	63230	887008	88755	8	747	66299985
	Ours (AC-based)	13410	4092765	19007	8	8112	154184784
283	[JSB+25]	113003	1740018	133832	11	3020	404172640
	Ours (Karatsuba)	113003	1545546	154378	9	1153	177997834
	Ours (AC-based)	18348	6768098	25303	9	11075	280230725
571	[JSB+25]	405223	6986368	468137	13	5815	2722216655
	Ours (Karatsuba)	405223	6047276	530478	10	2298	1219038444
	Ours (AC-based)	46397	34891194	61244	10	24406	1494721064

For the AC-based construction, let $m = \text{HW}(n-1) + \lfloor \log_2(n-1) \rfloor - 1$ and let l be the Toffoli-count of one multiplication. The corrected width is $n + 4l + ml = n + (m+4)l$, as stated in Theorem 2.